

Question 1

$$\Sigma F_y = F_g \perp - F_N = 0$$

$$F_N = F_g \perp$$

$$F_N = mg \cos \theta \dots (1)$$

$$\Sigma F_x = m \cdot a_x$$

$$0 = F_g \parallel - F_{fk}$$

$$mg \sin \theta = F_N \cdot \mu_k$$

$$F_N = \frac{mg \sin \theta}{\mu_k} \dots (2)$$

(1) into (2)

$$mg \cos \theta = \frac{mg \sin \theta}{\mu_k}$$

$$\therefore \mu_k = \frac{\sin \theta}{\cos \theta}$$

$$\mu_k = \tan \theta$$

Question 1

(m)

(2m)

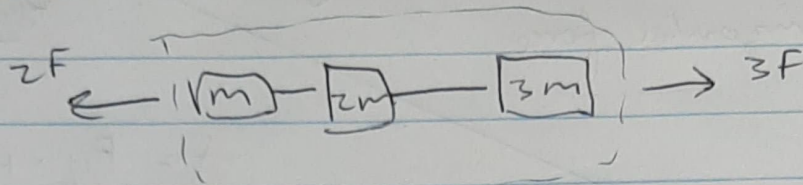
1.2)

$$F_{\text{net}} = m \cdot a$$

$$T_2 - 24 = ma$$

$$F_{\text{net}} = m \cdot a$$

$$T_1 - T_2 = 2ma$$



1.2) Tension = $3F - 2F$
 $= 3(12) + 2(12)$
 $= 60 \text{ N to the right}$

Question 2

2.1) $F_{\text{net}} = m \cdot a$

$$F_c - F_f = 0$$

$$F_c = F_f$$

$$\frac{m v^2}{R} = m(9,8)(0,25)$$

$$\frac{(33,33)^2}{300} = (9,8)(0,25)$$

$$3,7 \neq 2,45$$

$$F_N = F_g$$

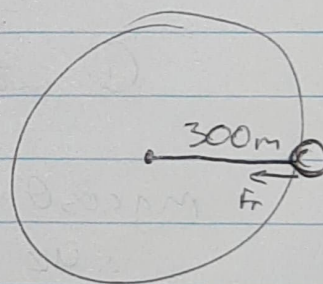
$$F_f = F_N \times \mu_k$$

=

$\therefore F_c$ (force pulling the car out of loop/circle)

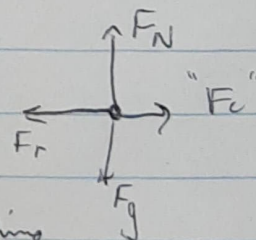
> Friction

force (force keeping car in circular path)



$$\mu = 0,25$$

$$v = 120 \text{ km/h} = 33,33 \text{ m/s}$$



$\therefore$ The car will veer towards the outside of the circle.

$$2.2) \quad F_{\text{net}} = m \cdot a$$

$$\therefore F_c = F_g + T$$

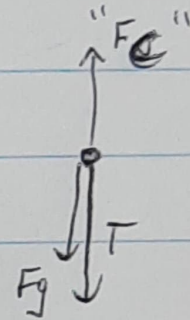
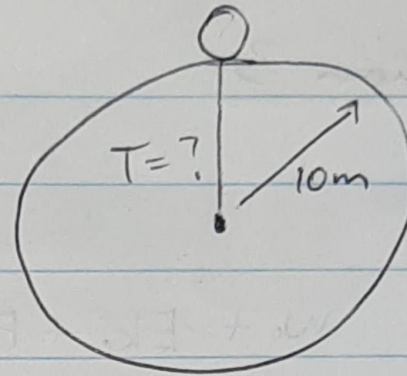
$$\frac{m v^2}{R} = \cancel{F_g} + T$$

$$\frac{(510,2)(5)^2}{10} = 5000 + T$$

$$\therefore T = -3724,5 \text{ N}$$

$$5 \text{ m/s}$$

$$F_g = 5000 \text{ N}$$



$$F_g = m g$$

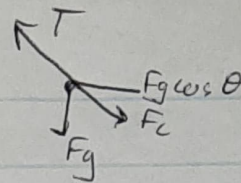
$$5000 = m (9,8)$$

$$m = 510,20 \text{ Kg}$$

$$\therefore \text{Force of Boom on the car} = 3724,5 \text{ N}$$

$$= 3,7 \text{ kN}$$

$$\begin{aligned}
 2.3) F_{\text{net}} &= T - F_g \cos \theta \\
 &= 8 - (0,3)(9,8) \cos 50 \\
 \therefore F_{\text{net}} &= 6,1 \text{ N}
 \end{aligned}$$



Question 3

$$\begin{aligned}
 3.1) W &= \int_{x=0}^{x=3} (fx) \cdot dx \\
 &= \int_0^3 (6x - 12) \cdot dx \\
 &= (3x^2 - 12x) \Big|_0^3
 \end{aligned}$$

$$\therefore W = -29 \text{ J}$$

$$\therefore E_p \text{ at } x=3 = -29 \text{ J}$$

$$\begin{aligned}
 3.2) a) P_x &= U + K \\
 &= -\frac{1}{2} kx^2 + \frac{1}{2} mU^2 \\
 U^2 &= \frac{P_x + \frac{1}{2} kx^2}{\frac{1}{2} m}
 \end{aligned}$$

$$\therefore U = \sqrt{\frac{P_x + \frac{1}{2} kx^2}{\frac{1}{2} m}}$$

$$\begin{aligned}
 b) U &= \sqrt{\frac{(80)(0,13) + \frac{1}{2}(800)(0,13)}{\frac{1}{2}(12)}} \\
 &= 3,2 \text{ m/s}
 \end{aligned}$$